Unit: Unit 3: Functions
Topic: Operations with Functions

Name: ________________
Date: __________

Foundation (Jalapeno)

Q1. If $f(x) = x^2$ and $g(x) = 2x$, what is $(f + g)(x)$?
(A) $x^2 + 2x$
(B) $x^2 - 2x$
(C) $2x^2$
(D) $2x^2 + x$
(E) $x^2 - x$

Q2. Given $f(x) = 1\frac{}{x}$ and $g(x) = 1\frac{}{x+1}$, what is the domain of $(f - g)(x)$?
(A) $(-\infty, -1) \cup (-1, \infty)$
(B) $(-\infty, -1) \cup (-1, 0) \cup (0, \infty)$
(C) $(-\infty, 0) \cup (0, \infty)$
(D) $(-\infty, -1)$
(E) $(-1, \infty)$

Q3. A basketball player's score can be represented by the function $s(x) = 2x$, where x is the number of shots. If the player's rival scores $r(x) = x + 5$, what is $(s - r)(x)$?
(A) $x - 5$
(B) $x + 5$
(C) $2x - 5$
(D) $2x + 5$
(E) $x - 10$

Q4. Given $f(x) = x^3$ and $g(x) = 2x$, what is $(f \cdot g)(x)$?
(A) $2x^4$
(B) $2x^3$
(C) $x^3 + 2x$
(D) $x^3 - 2x$
(E) $x^2 + 2x$

Q5. A cyclist travels a distance represented by the function $d(x) = 1\frac{}{2}\mathrm{x}^2$, where x is the time. If the cyclist's friend travels $f(x) = x + 2$, what is $(d - f)(x)$?
(A) $\frac{1}{2}x^2 - x - 2$
(B) $\frac{1}{2}x^2 + x + 2$
(C) $\frac{1}{2}x^2 - x + 2$
(D) $x^2 - x - 2$
(E) $x^2 + x + 2$

Q6. If $f(x) = x - 2$ and $g(x) = 1\frac{}{x+1}$, what is $(f \cdot g)(x)$?
(A) $\frac{x-2}{x+1}$
(B) $\frac{x+2}{x-1}$
(C) $\frac{x+1}{x-2}$
(D) $\frac{2-x}{x+1}$
(E) $\frac{x-1}{x+2}$

Q7. Given $f(x) = x^2 + 1$ and $g(x) = x - 1$, what is $(f/g)(x)$?
(A) $\frac{x^2+1}{x-1}$
(B) $\frac{x^2-1}{x+1}$
(C) $\frac{x+1}{x-1}$
(D) $\frac{x-1}{x+1}$
(E) $\frac{x^2+1}{x+1}$

Q8. A tennis player's ranking can be represented by the function $r(x) = 3x - 2$, where x is the number of wins. If the player's opponent's ranking is $o(x) = x + 4$, what is $(r - o)(x)$?
(A) $2x - 6$
(B) $2x + 6$
(C) $3x - 4$
(D) $3x + 4$
(E) $x - 6$

Open Ended Questions (FRQ)

Q9. Given $f(x) = x^2 - 2$ and $g(x) = x + 3$, find $(f + g)(x)$ and determine its domain. Show all work.

Q10. A runner's distance can be represented by the function $d(x) = 1\frac{}{3}\mathrm{x}^2$, where x is the time. If the runner's friend's distance is $f(x) = 2x - 1$, find $(d - f)(x)$ and simplify. Show all work.

Chef's Tips

- Remind students to simplify expressions after operations.
- Caution students to consider the domain when combining functions.
- Encourage students to check their work by plugging in values.